

fastDiversity: an R package for population genetic diversity analysis of multi-species SNP datasets

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Software

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Summary

Genetic diversity underpins evolutionary potential, adaptive capacity, and population persistence, making its quantification central to conservation genetics and population biology (Frankham et al., 2010). Reduced-representation sequencing approaches including DArTseq (Sansaloni et al., 2011) and RADseq (Baird et al., 2008) now routinely generate large biallelic single nucleotide polymorphism (SNP) datasets spanning hundreds of individuals and thousands of loci. Increasingly, these datasets include multiple closely related species or divergent lineages analysed within a comparative framework.

fastDiversity is an R package for rapid calculation of population genetic diversity statistics from biallelic SNP dosage matrices. The package operates directly on standard numeric genotype matrices encoded as 0, 1, 2, and NA, avoiding conversion to specialised genetic data classes. fastDiversity implements allele-sharing and private allele analyses; site- and population-level heterozygosity and inbreeding statistics; Weir–Cockerham F-statistics (Weir & Cockerham, 1984); nucleotide diversity, Watterson's theta, and Tajima's D; multilocus heterozygosity; and Hardy–Weinberg equilibrium testing with optional bootstrap confidence intervals.

A defining feature of the package is that locus-level filtering by minor allele frequency, missingness, and allele count is performed independently within user-defined genetic groups. This enables biologically appropriate analysis of multi-species datasets within a single workflow, avoiding artefacts introduced by pooled filtering across reproductively isolated groups.

fastDiversity has been under active development since 2024 and the software is archived on Zenodo ([doi:10.5281/zenodo.12741582](https://doi.org/10.5281/zenodo.12741582)).

Statement of need

Several R packages provide population genetic analysis tools for biallelic SNP datasets, including hierfstat (Goudet, 2005), diveRsity (Keenan et al., 2013), adegenet (Jombart, 2008), dartR (Gruber et al., 2018), and poppr (Kamvar et al., 2014), collectively covering diversity statistics, ordination, and population structure inference. These tools are widely used but generally apply locus-level filtering globally across the pooled dataset, or require datasets to be manually split by species prior to analysis.

This presents a recurring problem for multi-species datasets. When MAF thresholds, missingness filters, or expected heterozygosity are calculated across pooled samples from divergent taxa, allele frequencies reflect a mixture of reproductively isolated gene pools rather than any biologically meaningful unit. As a result, loci that are polymorphic within a species may be incorrectly removed by pooled MAF filters, while loci that are fixed within species but differentiated between them can inflate diversity or F_{IS} estimates when treated as within-population variation. Differences in locus presence and missingness across taxa further bias locus retention under standard workflows (Table 1). Species-specific filtering avoids these

41 artefacts by evaluating loci within genetic groups, preserving within-taxon signal while enabling
42 comparable summaries across taxa.
43 Beyond filtering correctness, performing independent analyses per genetic group simultaneously,
44 rather than through repeated manual subsetting, reduces friction and allows comparative
45 diversity analyses to sit alongside other standard steps such as dimensionality reduction,
46 ancestry estimation, and pairwise FST within a single analysis script.
47 fastDiversity was developed to address this gap. It applies filtering and diversity estimation
48 independently within user-defined genetic groups while returning results in a unified tidy
49 output, enabling scalable multi-species analyses without repetitive manual re-analysis. The
50 package operates directly on standard SNP dosage matrices rather than requiring conversion
51 to specialised object classes such as genind or genlight, lowering the barrier for users working
52 within existing pipelines. Show Image

Table with 6 columns: Species, Locus 1, Locus 2, Locus 3, Locus 4, Locus 5. Rows show genotype data for species A, B, and C across five loci, with values 0, 1, 2, or NA.

Figure 1: Table 1. Effects of pooled vs species-specific filtering in multi-species SNP datasets. Example genotype matrix showing how locus informativeness differs across three species (A–C) under pooled filtering. Loci may be informative across all species (L1), polymorphic only within one species and potentially removed by pooled MAF filters (L2), alternatively fixed within species inflating pooled summary statistics such as FIS (L3), informative in one species but fixed in others and potentially removed by pooled MAF filters (L4), or present only in a subset of species and excluded under pooled missingness filtering (L5). Species-specific filtering retains different informative loci per taxon (A: L1, L4; B: L1, L5; C: L1, L2), illustrating how pooled filtering can distort comparative SNP datasets.

53 Software design and functionality

54 fastDiversity is designed around two principles: biologically appropriate filtering and minimal
55 data-handling overhead.
56 The package accepts standard biallelic SNP dosage matrices produced by common variant-
57 calling pipelines, together with plain metadata vectors describing sample groupings. This design
58 allows integration into existing R analysis pipelines without object conversion or restructuring.
59 All locus-level quality filters, including missingness and minor allele frequency thresholds,
60 are applied independently within each user-defined genetic group. This filtering framework
61 propagates throughout the package and ensures that downstream diversity statistics are
62 calculated only from loci informative within the relevant biological unit.
63 Implemented analyses span multiple biological scales, including allele-level summaries such as
64 allele sharing and private alleles; population-level diversity statistics including observed and
65 expected heterozygosity, inbreeding coefficients, and Weir–Cockerham F-statistics (Weir &
66 Cockerham, 1984); sequence diversity statistics including nucleotide diversity, Watterson’s theta,

and Tajima's D using missing-data-aware approaches (Korunes & Samuk, 2021); individual-level summaries including multilocus heterozygosity and Wright's inbreeding coefficient. Where appropriate, bootstrap and resampling procedures adapted from Goudet (2005) and Keenan et al. (2013) are implemented to estimate confidence intervals and standardise comparisons among uneven sample sizes.

All functions return tidy tabular outputs compatible with standard R visualisation and downstream analysis workflows.

State of the field

fastDiversity complements rather than replaces existing population genetic software ecosystems. hierfstat (Goudet, 2005) and diveRsity (Keenan et al., 2013) provide established implementations of diversity and differentiation statistics but apply filtering globally and require specialised input structures. adegenet (Jombart, 2008) and dartR (Gruber et al., 2018) provide extensive infrastructure for genomic analyses and DArTseq workflows but are centred on custom object classes and single-species analytical assumptions. poppr (Kamvar et al., 2014) supports clonal and mixed-ploidy systems but does not specifically address per-group filtering in comparative multi-species datasets.

fastDiversity occupies a complementary niche focused on lightweight analysis of comparative SNP datasets in which biologically scoped filtering is required across many taxa or lineages simultaneously.

Research impact statement

fastDiversity has been used in conservation genomics analyses of threatened Australian flora. McMaster, Pemberton, et al. (2025) used the package to compare diversity and population structure across multiple species of Homoranthus (Myrtaceae). McMaster, Duretto, et al. (2025) applied the package in a conservation genomic assessment of the critically endangered Zieria obcordata (Rutaceae), identifying extremely low within-population diversity and informing subspecies delimitation. The package has additionally been used in over 30 confidential threatened-species assessments conducted for the New South Wales Government Saving our Species program, demonstrating applicability in operational conservation management contexts.

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AI usage disclosure

Claude Sonnet 4.6 (Anthropic) and ChatGPT (OpenAI GPT-5.5) were used to assist with editing and restructuring documentation and manuscript text based on author-provided drafts and outlines. All scientific content, software design decisions, implementations, analyses, and interpretations were developed, verified, and approved by the author.

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